SPARKY'S SCIENCE EXPERIMENTS

Floating Needle Compass

Natural Science (magnetism) · Ages 8+ · 15 minutes

Make your own compass: turn a sewing needle into a magnet and float it on water so it points north and south.

Safety first

Sparky always says: Keep magnets away from little children - swallowing them is very dangerous!

- Needles are sharp - an adult should help.
- Keep magnets away from phones and bank cards.

You need

- A steel sewing needle (or a straightened paper clip)
- A strong magnet, e.g. from an old speaker (fridge magnets are usually too weak)
- A bottle cap, piece of cork or small leaf
- A bowl of water

Steps

1. Stroke the needle with one end of the magnet, from the eye to the point, 30 to 40 times. Always the same way: lift the magnet off between strokes.
2. Lay the needle on the bottle cap, cork or leaf.
3. Float it in the middle of the bowl, away from metal and phones.
4. Watch it turn and settle. Push it gently - it swings back to the same line!
5. Which end is north? At midday in South Africa the sun is in the northern sky, so the end pointing towards the sun's side is north.

If it doesn't work, try these in order

1. Doesn't turn? Move away from metal and phones, and keep the float off the side of the bowl.
2. Points a different way each time? Stroke it more, always the same way.

Why it works

Earth is like a giant magnet. Stroking the needle lines up tiny magnetic areas in the steel, so it becomes a small magnet with a north and a south pole. Floating lets it turn easily, and Earth's magnetism pulls it into a north-south line. In South Africa a compass points about 18 to 30 degrees west of true north: about 21 in Johannesburg, 27 in Cape Town and Durban.

Did you know? At night, the Southern Cross stars help you find south: follow the long arm of the cross about four and a half times its length, then look straight down to the horizon.

Ask Sparky: What happens if you bring the magnet close? Why keep a compass away from metal?